Gabriel Franco

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Summary

- Machine learning student and research assistant with a background from ENS Paris-Saclay (Computer Science & Electrical Engineering) and the IASD MSc (ENS Ulm / Mines Paris / Dauphine-PSL). I work on geometric and topological deep learning, computer vision and applied machine learning, with hands-on experience across research labs, public administration and industrial R&D. Admitted to the French Agrégation in Engineering and Computer Sciences (national rank 15/151).

Experience

INRIA Saclay & CentraleSupélec (MICS Lab), Research Assistant — Geometric & Topological Deep Learning
Gif-sur-Yvette, France
Mar 2026 – Aug 2026
6 months

- Developing methods to tackle oversmoothing and oversquashing in graph learning through rewiring and lifting techniques.
- Working with topological and graph neural network architectures (CWN, GIN, GPS, GAN, CIN) and recent topological deep learning tooling (DiffLift, TopoTune, TopoModelX).

Ministry of the Economy and Finance — Directorate for Tax Studies and Statistics, Data Scientist Intern — Tax Fraud Estimation
Paris, France
Oct 2025 – Feb 2026
5 months

- Developed data-science models to evaluate tax fraud and under-declaration using ensemble learning (XGBoost).
- Improved the model F1-score from 0.16 to 0.23 with Bayesian hyperparameter optimization.
- Ran explainable-AI analysis using Shapley values on gradient-boosted trees to interpret predictions.

French Embassy in Berlin — Service for Science and Technology, Intern — Scientific Cooperation
Berlin, Germany
June 2025 – Sept 2025
4 months

- Coordinated the first "Fête de la Science" in Germany — 20 events across 7 Franco-German high schools (communication, management, logistics).
- Conducted strategic mapping of the German scientific-culture and innovation ecosystem for diplomatic reporting.

Hublex, R&D Part-time Engineer — RNN Control for an Exoskeleton
Cachan, France
Sept 2024 – June 2025
10 months

- Designed an industrial dossier on the neural-network control of an exoskeleton.
- Implemented and tuned an LSTM (RNN) command model from sensor inputs, focusing on data manipulation and Edge-AI deployment — projected 50%+ increase in adoption rate for industrial clients.

Università di Pavia, Deep Learning Project Intern — Post-seismic Damage Detection
Pavia, Italy
June 2024 – Aug 2024
3 months

- Developed and fine-tuned a damage-detection algorithm with the YOLOv8/v9/v10 frameworks on post-seismic images, improving mean Average Precision for damage localization by 0.16.
- Gained hands-on experience with ultralytics, PyTorch and TensorFlow.

ENS Paris-Saclay, Autonomous Racing Car — COVAPSY Competition
Paris-Saclay, France
Jan 2024 – June 2024
6 months

- Computer vision and reinforcement learning for guiding a reduced-scale autonomous car.
- Developed vision-based spatial reasoning for autonomous navigation.

Les Glénans, Catamaran Instructor — France
- Sailing instructor at Les Glénans, France's largest sailing school, since 2020. — Jan 2020 – present, 6 years 9 months

Education

MSc **ENS Ulm, Mines Paris & Université Paris Dauphine-PSL**, Artificial Intelligence, Machine Learning and Data Science (IASD) — Paris, France, Sept 2026 – June 2027
- Coursework: Reinforcement Learning, Theoretical Foundations of Machine Learning, Deep Learning for Image Analysis, Kernel Methods, Robot Learning, Dynamical Systems in ML, Bayesian Machine Learning.

MSc **ENS Paris-Saclay (Université Paris-Saclay)**, Computer Science and Electrical Engineering — Gif-sur-Yvette, France, Sept 2023 – June 2025
- Coursework: Reinforcement Learning, Audio/Image/Signal Processing, Deep Learning & AI (CNN & Bayesian NN), Inverse Problems, HPC, Databases & SQL, Information Theory & Source Coding (incl. VAE), Embedded Systems & Edge Computing, Operating Systems, Networking, Wavelets, Software Engineering.
- Admitted to the Agrégation (national teaching competition for higher-education professors) in Engineering and Computer Sciences — nationwide rank 15/151.

BSc **ENS Paris-Saclay (Université Paris-Saclay)**, Engineering — Gif-sur-Yvette, France, Sept 2022 – June 2023
- Coursework: Object-Oriented Programming, Advanced Probability & Stochastic Processes, Estimation & Optimization Theory, Signal Processing, Linear Systems & Control Theory.

CPGE **Lycée Sainte-Geneviève / Lycée de Cachan**, Preparatory Classes for the Grandes Écoles (PTSI / PT*, Advanced Track) — Versailles / Cachan, France, Sept 2019 – June 2022
- Advanced Mathematics, Engineering Sciences and Physics.
- Admitted to ENS Paris-Saclay — nationwide rank 188/2560.

Projects

Local Edge-AI pipeline on Radxa Rock 4D — Jan 2025 – present

On-device LLMs and VLM-based vision tools for real-time spatial analysis on a single-board computer, with no cloud dependency.
- Quantized LLM / VLM inference running fully on-device.
- Multiple companion projects on Arduino Uno; regular hackathon participation.

Technical Skills

Programming: Python (NumPy, SciPy, OpenCV, Matplotlib), C, C++, MATLAB, SQL, VHDL

AI / ML: PyTorch, PyTorch Geometric, Keras, ultralytics, TensorFlow, scikit-learn, CUDA, Apache Spark, Pandas

Tools: Git / GitHub, Bash & Unix, LaTeX, Typst, Notion, MS Office, Canva

Language Skills

French: Native

English: Professional working proficiency (IELTS 7 — C1)

German: Professional working proficiency (B2–C1)

Italian: Conversational (B1)

Interests & Activities

Sports: Triathlon (x2 Half Ironman), catamaran sailing and instructing

Music: Flute — 16 years of conservatory, symphony and jazz orchestras

Teaching: Maths & Physics/Chemistry tutoring; four guest CS lectures at MSc level

Academic: L3-level Econometrics & Macroeconomics; Philosophy of Science and Theory of Knowledge; politics